

Data Visualization Using PowerBi

Procedure:

1. Open Power BI Desktop & Inspect Interface

1. Launch Power BI Desktop.
2. Note key panes: Fields, Visualizations, Filters, Data/Report/Model view.
3. Switch to Data view briefly to understand where tables and columns appear.

2. Connect to Data Sources

****A. Excel****

1. Home → Get data → Excel → choose `SalesData.xlsx`.
2. In Navigator, check the Sales sheet and click Load.

****B. CSV****

1. Home → Get data → Text/CSV → choose `Targets.csv`.
2. Preview and click Load.

3. Clean & Transform (Power Query)

1. Home → Transform data to open Power Query Editor.
2. For Sales table:
 - Ensure Date column has Date datatype.
 - Create a TotalSales column: Quantity * UnitPrice (Add Column → Custom Column → `=[Quantity]\*[UnitPrice]`).
 - Trim whitespace in text columns; remove duplicates if needed.
3. For `Targets.csv`:
 - Convert Month to date or text format consistently (e.g., YYYY-MM).
4. Close & Apply.

4. Create Basic Visualizations

Create a new report page and add visualizations:

1. ****Bar chart**** — Sales by Category
 - Axis: Category
 - Value: Total Sales
 - Add data labels, sort descending.
2. ****Line chart**** — Monthly Sales Trend
 - Axis: Sales[YearMonth] (set type to categorical or continuous as needed)
 - Values: Total Sales
 - Add a slicer for Region.
3. ****Pie chart / Donut**** — Sales proportion by Region
 - Legend: Region
 - Values: Total Sales
4. ****Card visuals**** — Key KPIs
 - Cards showing Total Sales, Total Quantity, Avg Order Value.
5. ****Table / Matrix**** — Top 10 Products by Sales
 - Columns: Product, Total Sales, Total Quantity

- Use Top N filter on Product to show top 10.

5. Build a Dashboard-like Report Page

1. Arrange visuals into a clean layout:

- Top row: KPI cards and a date/region slicer.
- Left column: bar chart (Category) & table (Top products).
- Right column: line chart (trend) & pie/donut (region share).
- Bottom: gauge or variance chart comparing sales vs target.

2. Add titles, tooltips, and consistent formatting (font sizes, number formats).

3. Save the `.pbix` file.

Output:

